COMPUTER VISION - LAB 4

Computer Vision 2024-25
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Topics: Detect road signs and traffic lights in driving scenes.
Goal: Find all road signs and classify them, find traffic lights and get their status.

Write a Python script which includes methods that:

1. Load some of the provided images
    - There are 2 sets: an "easy" one where the detection is relatively simple and an "hard" one with more challenging images.
    - Links to large datasets are also provided, use them in case you decide to train ML algorithms or need a large amount of data to tune your approach.
2. Find the road signs and the traffic lights:
    - You can use Deep Learning based object detectors e.g. Yolo, R-CNN, SSD, Retina-net, etc..
    - Or classic ML approaches, e.g., Bag-of-Words or Viola and Jones
    - Also "deterministic" strategies e.g., Hough Transform can be used
3. For each traffic light analyze it to get if it is green, yellow or red (e.g., you can analyze the color in the different up-mid-low regions or use a ML classifier or analyze the color histogram)
4. For each speed limit sign read the maximum allowed speed value (you can find the central region and use some OCR approach or template matching)

Optional steps

1. Try different schemes for detecting or classifying road signs and traffic light, e.g., Hough Transform or template matching
2. Generalize to different types and colors of the signs or to different types of non-standard traffic lights
3. Compare classification and OCR for reading speed limits
4. Recognize more types of road signs
5. Add provisions for night/dark environments
6. Test other images
7. Any additional idea is welcome!